



DAYANANDA SAGAR
UNIVERSITY



SCHOOL OF
ENGINEERING

School of Engineering
Department of Computer Science and Engineering

B Tech, Minor Project(2025-26)

Names & USN of Team Members	Bhavish S Shetty	ENG23CS0035
	Bhaskar A Gowda	ENG23CS0032
	Ranjith Shetty	ENG23CS0424
	Basavanagouda Meti	ENG23CS0029
Name of the Guide	Prof. Yashpal Gupta S	
Signature of the Guide		
Final Project Title	Predictive Modelling of Leading Indian Banks' Stock Prices using NLP, Deep Learning & FinBERT	
Team No	35	

ABSTRACT

The Indian stock market plays a crucial role in the country's economic growth, with the banking sector being one of its most influential components. Predicting bank stock prices is challenging due to the complex interaction between numerical market indicators and investor sentiment influenced by financial news. This project proposes a predictive modelling framework for forecasting stock prices of top Indian banks by integrating historical stock price data with sentiment information extracted from financial news using Natural Language Processing (NLP) techniques. A domain-specific sentiment analysis model is used to generate positive, negative, and neutral sentiment scores, which are combined with historical price attributes such as open, high, low, close prices, and trading volume. Deep learning models, including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, are employed to capture temporal dependencies and non-linear patterns in the data. Model performance is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score. Experimental results show that incorporating sentiment-based features improves prediction accuracy compared to models using only historical price data, demonstrating the effectiveness of combining NLP and deep learning approaches for stock price forecasting in the Indian banking sector.



Guide Signature